

EE2213 Introduction to Artificial Intelligence

Tutorial 1

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Common questions and clarifications

Q: Is today tutorial?

A: This is a *packaged lecture + tutorial* course. Some lectures are on Fridays, some tutorials on Mondays. You can always check the **schedule file** on Canvas.

Q: Difference between EE2211 and EE2213?

A: EE2213 covers broader AI concepts, including machine learning, with more focus on implementation and applications.

Q: AI models vs. AI agents?

A: AI models can be part of an agent; they are the *computational components* that map inputs to outputs for the agent.

Q: What's required to prepare for the tutorial?

A: At a minimum, you should read through the questions before class.

Q1 Which of the following statements about agents are true? Select all that apply.

- a) Human and robot agents differ because the latter do not have actuators.
- b) Eyes and ears are examples of a human agent's actuators.
- c) An agent senses its environment through perceptors.
- d) Percepts are the agent's perceptual inputs from the environment at any given time.
- e) Touchless faucets can also be viewed as agents.

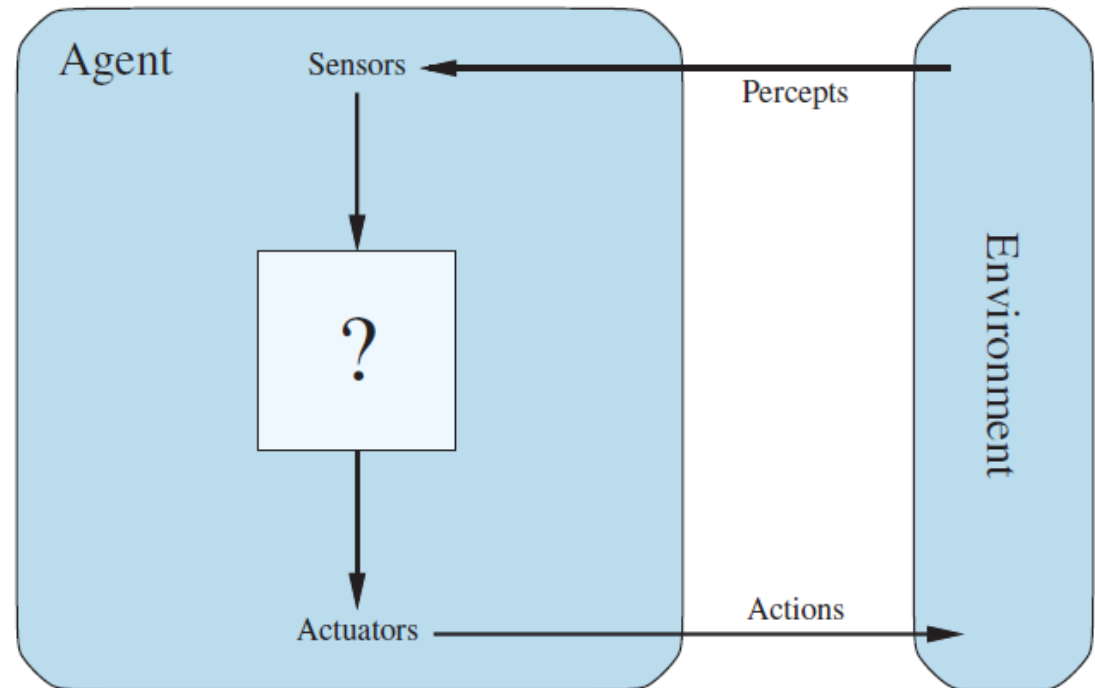
Recap: Definition of Agent

An **agent** is anything that **perceives** its environment through **sensors** and can **act** on its environment through **actuators**

A **percept** is the agent's **perceptual inputs** at any given instance

An **actuator** is the part of an agent that allows it to take **action** in the environment based on its decisions.

A **sensor** is a device or mechanism that allows an agent to perceive and gather information about its environment.



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P54 – P57.

Q1 Which of the following statements about agents are true? Select all that apply.

- a) Human and robot agents differ because the latter do not ~~have~~ actuators. In AI, all agents have actuators, which are necessary for action.
- b) Eyes and ears are examples of a human agent's ~~actuators~~. ← sensors
- c) An agent senses its environment through ~~perceptors~~. ← sensors
- d) Percepts are the agent's perceptual inputs from the environment at any given time. ✓
- e) Touchless faucets can also be viewed as agents. ✓

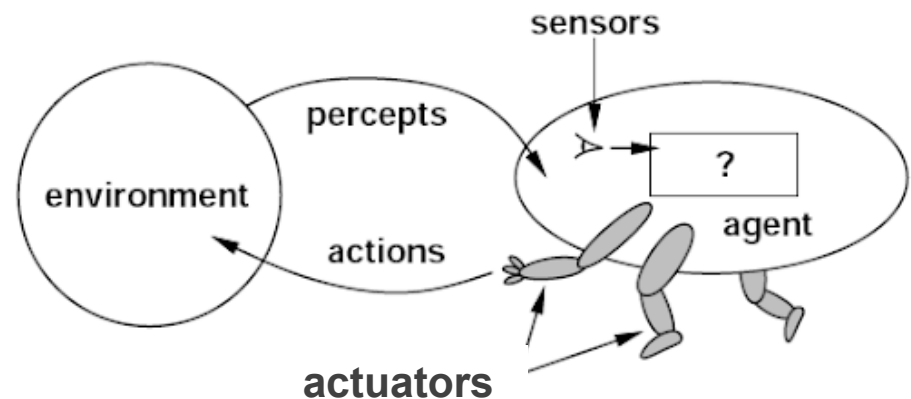
Q2 Which of the following statements regarding rational agents are true? Select all that apply.

- a) Rational agents always know the environment dynamics and can predict future updates of the environment.
- b) A rational agent is omniscient
- c) A rational agent selects an action that maximizes the expected value of its performance measure.
- d) A rational agent is not guaranteed to find optimal solutions.

Recap: Definition of Rational Agent

Rational Agent:

- For each possible percept sequence P
- Select an action a **expected** to **maximize** the **performance measure**
- Base decision on:
 - *Evidence from the percept sequence*
 - *Built-in knowledge of the agent*



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P58.

Definition of Rational Agent (cont.)

- Are rational agents **omniscient**?

No – they only know what their sensors tell them.

- Are rational agents **clairvoyant**?

No – they may lack knowledge of the environment dynamics. They don't magically know the future or hidden parts of the environment.

- Do rational agents **explore** and **learn**?

Yes – in unknown environments these are essential

- Are rational agents **autonomous** (i.e., transcend initial program)?

Yes – as they learn, their behavior depends more on their own experience

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P58 – P60.

Q2 Which of the following statements regarding rational agents are true? Select all that apply.

- a) Rational agents always ~~know~~ the environment dynamics and can predict future updates of the environment. Agents are not clairvoyant
- b) A rational agent is ~~omniscient~~
- c) A rational agent selects an action that maximizes the expected value of its performance measure ✓
- d) A rational agent is not guaranteed to find optimal solutions. ✓

Two levels of optimality:

- *Actual optimality*: what truly turns out to be best
- *Agent-optimality (rationality)*: what is best according to the agent's knowledge at the time

Q2 Takeaway: Understanding rationality



A **rational** agent is *not guaranteed* to find optimal solutions

With *perfect* knowledge and *unlimited* computing power, the agent would find the true optimal solution.

However, the agent is **not omniscient or clairvoyant**: it does **not have perfect knowledge** of the environment.

In reality, agents have incomplete or noisy information and **act optimally based** on the **current evidence** available—this is called ***bounded rationality***.

The key distinction:

- Actual optimality: the best possible outcome in reality.
- **Rationality**: the best **expected** outcome based on the agent's **current evidence and existing knowledge**.

Q3. Which of the following are necessary to formally define a problem for a problem-solving agent? Select all that apply.

- a) A clear goal condition that defines when the agent has successfully solved the problem
- b) Specifying the outcome of each action that the agent could take
- c) The initial state from which the agent begins its problem-solving process
- d) The set of possible actions the agent can perform in each state

Definition of Problem Solving Agents

- **Adopt a Goal:** The agent has a specific objective to achieve.
- **Plan Ahead:** The agent evaluates possible *future actions* and their outcomes, typically by examining the outcomes of each available action in every state and assessing their desirability.
- **Find a Solution:** The agent determines *a sequence of actions* that will lead it from the *initial state* to the *goal*.

Q3. Which of the following are necessary to formally define a problem for a problem-solving agent? Select all that apply.

- a) A clear goal condition that defines when the agent has successfully solved the problem ✓
- b) Specifying the outcome of each action that the agent could take ✓
- c) The initial state from which the agent begins its problem-solving process ✓
- d) The set of possible actions the agent can perform in each state ✓

